



CardioIQ

Knowledge Graph™

A Semantic Intelligence Engine for
Precision Cardiovascular Care

ABSTRACT

CardioIQ Knowledge Graph™ is the semantic intelligence engine behind the CardioIQ Clinical Intelligence Platform. It organizes cardiovascular medicine as a connected network of biomedical knowledge — linking biomarkers, diseases, guidelines, therapies, imaging, lifestyle factors, and patient characteristics through clinically meaningful relationships.

Unlike conventional decision-support tools that apply isolated rules or black-box models, CardioIQ enables explainable, evidence-based reasoning. Every recommendation is traceable to activated graph nodes, traversed relationships, supporting evidence (PMID/DOI), and specific guideline statements — revealing hidden risk patterns that standard lipid panels cannot detect.

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Prepared By	Tea Gamezardashvili, MD, PhD, FACC, MHA
Role	Preventive Cardiologist · Lipidologist · Founder, CardioIQ
Organisation	CardioIQ Health™ · cardioiq.health
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Executive Summary

CardioIQ Knowledge Graph™ is the semantic intelligence engine behind the CardioIQ Clinical Intelligence Platform. It organizes cardiovascular medicine as a connected network of biomedical knowledge — linking biomarkers, diseases, guidelines, therapies, imaging, lifestyle factors, and patient characteristics through clinically meaningful relationships.

Unlike conventional decision-support tools that apply isolated rules or black-box models, CardioIQ enables explainable, evidence-based reasoning. Every recommendation is traceable to activated graph nodes, traversed relationships, supporting evidence (PMID/DOI), and specific guideline statements — revealing hidden risk patterns that standard lipid panels cannot detect.

The Problem: Fragmented, Unexplainable Decision Support

Modern cardiovascular care faces three critical challenges:

Information Overload	Clinicians must synthesize data from labs, imaging, comorbidities, and rapidly evolving guidelines simultaneously.
Hidden Residual Risk	Patients often remain at high risk despite "on-target" traditional metrics (e.g., LDL-C). ApoB, Lp(a), inflammatory burden, and insulin resistance remain undetected.
Black-Box AI	Many AI-driven tools cannot explain why a recommendation was made, limiting clinical trust and institutional adoption.

CardioIQ addresses these challenges by representing cardiovascular medicine as a multi-layered, evidence-anchored Knowledge Graph that supports deterministic, auditable reasoning — where every clinical recommendation is traceable to activated graph nodes, traversed relationships, and specific guideline statements.

Our Vision

To organize the world's cardiovascular knowledge into a structured, queryable, and explainable semantic network that:

- Reveals hidden risk patterns not visible on standard lipid panels
- Aligns every recommendation with current international guidelines
- Makes AI reasoning transparent and clinically trustworthy

01 Design Principles

CardioIQ Knowledge Graph™ is built on five core principles:

1. Biological Accuracy

Relationships reflect accepted human pathophysiology and established biochemical mechanisms.

2. Clinical Validity

Every edge is supported by peer-reviewed evidence or internationally recognised guideline recommendations.

3. Explainability

Every inference is traceable, reproducible, and human-understandable — decomposable into patient findings, graph paths, and source evidence.

4. Modularity

Domains can be expanded independently — lipids, inflammation, renal, metabolic, vascular ageing — without structural redesign.

5. Version Control

Every node and relationship includes creation date, version number, source reference, evidence level, guideline version, and confidence score.

02 High-Level Architecture

The CardioIQ platform processes clinical data through a structured eight-stage pipeline from raw patient data to the final Clinical Intelligence Report. The Knowledge Graph sits at the centre, connecting all clinical reasoning layers.

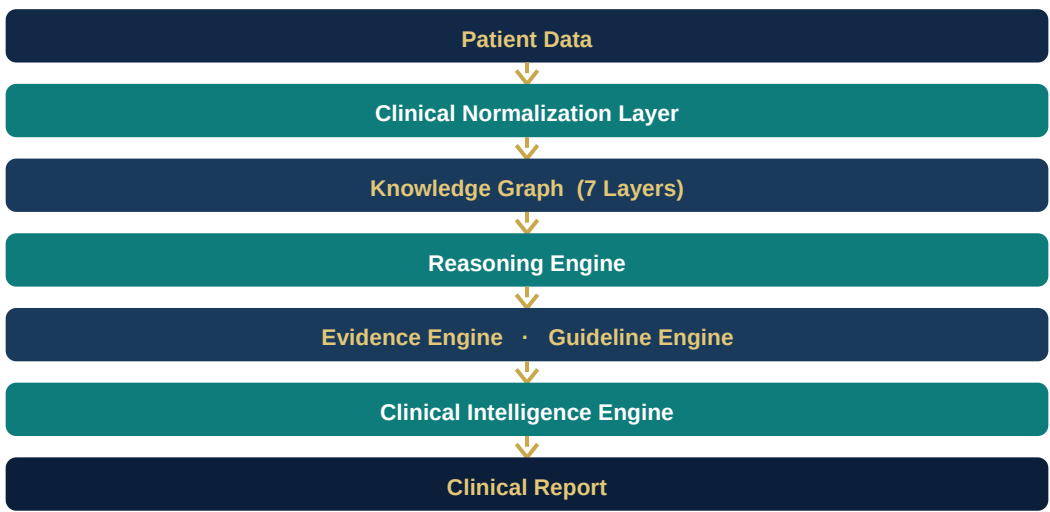


Diagram 1: CardioIQ Architecture Overview — eight-stage pipeline from patient data to clinical report.

03 Clinical Ontology: What CardioIQ "Understands"

CardioIQ recognizes and links concepts across seven key clinical domains. Each domain is independently expandable as the evidence base evolves:

Domain	Key Entities
Patient	Age · Sex · Ethnicity · Smoking · BMI · Exercise · Diet
Biomarkers	LDL-C · ApoB · Lp(a) · HDL-C · TG · hsCRP · HbA1c · Creatinine · eGFR · ACR
Diseases	ASCVD · FH · Hypertension · Diabetes · CKD · NAFLD · Obesity · HF · PAD · Stroke
Risk Factors	Smoking · Family History · Obesity · HTN · Inflammation · Sedentary Lifestyle · Sleep Disorders
Medications	Statins · Ezetimibe · PCSK9i · Inclisiran · Bempedoic Acid · GLP-1 RA · SGLT2i · ACEi · ARB
Imaging	CAC Score · Carotid Ultrasound · ABI · Echocardiography · CCTA
Guidelines	ESC · EAS · ACC · AHA · ADA · KDIGO · USPSTF · IAS

04 Core Relationship Types

Every connection in the Knowledge Graph is a semantic relationship with type, evidence metadata, and a confidence score:

ApoB	CAUSES	Atherosclerosis
CKD	AMPLIFIES	ASCVD Risk
Diabetes	ACCELERATES	Atherosclerosis
Lp(a)	INCREASES	Residual Risk
Guideline	RECOMMENDS	Therapy / Monitoring
Residual Risk	MODIFIES	Treatment Strategy

All Supported Relationship Types:

CAUSES	ASSOCIATED_WITH	PREDICTS	RECOMMENDS
TREATS	CONTRAINDICATED_IN	REQUIRES	MODIFIES
AMPLIFIES	SUPPRESSES	INTERACTS_WITH	DERIVED_FROM
SUPPORTED_BY	SUPERSEDES		

Canonical Example: ApoB **CAUSES** ASCVD | Source: ESC/EAS 2025 | Class I, Level A | Confidence: 0.99

This ensures every inference is scientifically grounded and auditable.

Node A	Relationship	Node B	Evidence Level
ApoB	CAUSES	Atherosclerosis	Class I / Level A
Atherosclerosis	LEADS_TO	ASCVD	Class I / Level A
Lp(a)	INCREASES	Residual Risk	EAS Consensus 2022
CKD	AMPLIFIES	ASCVD Risk	KDIGO 2022
Diabetes	ACCELERATES	Atherosclerosis	ESC 2023
PCSK9i	TREATS	Elevated ApoB	Class I / Level A
SGLT2i	TREATS	Residual CV Risk (T2DM)	Class I / Level A
Residual Risk	MODIFIES	Treatment Strategy	ESC/EAS 2019

05 Multi-Layer Graph Architecture

The Knowledge Graph employs seven interconnected semantic layers. Each layer is independently expandable. Layer 7 (Patient) is generated dynamically for each individual, creating a personalised clinical subgraph.

L1	Biomedical	Genes · Proteins · Biomarkers · Metabolic Pathways
L2	Disease	ASCVD · CKD · Diabetes · Obesity · HF · PAD · Stroke
L3	Risk	Absolute Risk · Lifetime Risk · Residual Risk · Inflammatory Risk
L4	Therapeutic	Lifestyle · Medication · Monitoring · Follow-up Protocol
L5	Guideline	ESC/EAS · ACC/AHA · ADA · KDIGO · USPSTF · IAS
L6	Evidence	RCTs · Meta-analyses · Consensus Statements · Registries
L7	Patient	Individual patient subgraph — personalised risk representation

06 Evidence Layer

Every relationship in the Knowledge Graph is anchored to structured evidence metadata. This ensures every recommendation is traceable to its source — not generated by a statistical model.

Field	Description	Example
PMID / DOI	Unique publication identifier	10.1093/eurheartj/ehz455
Guideline	Source guideline document	ESC/EAS 2019 Dyslipidaemias
Evidence Level	Class and level of evidence	Class I, Level A
Confidence Score	Graph-assigned confidence weight	0.99
Date	Publication or guideline year	2019 / Updated 2025
Version	Ontology version of this edge	v1.2

Example Edge: ApoB **CAUSES** Atherosclerosis | Source: ESC/EAS 2019 | Class I, Level A | Confidence: 0.99

07 Reasoning Engine

The CardioIQ Reasoning Engine performs deterministic, explainable inference through the Knowledge Graph. Unlike generative LLMs, the engine never invents intermediate conclusions — every inference corresponds to a graph relationship supported by documented evidence.

Example Inference Chain:

Input	LDL-C = 2.1 mmol/L (near-normal on statin therapy)
Detected	ApoB = 125 mg/dL (above very-high-risk target of 65 mg/dL)
Inference	LDL-C / ApoB Discordance — small dense LDL phenotype
Risk Domain	Residual Lipid Risk — atherogenic particle burden uncontrolled
Missing	Lp(a) not measured — independent genetic risk unquantified
Recommend	Measure Lp(a) + Intensify lipid-lowering therapy
Guideline	ESC/EAS 2019 — ApoB target <65 mg/dL (Class I, Level B)
Output	Clinical recommendation with evidence reference and priority level

08 Explainability Layer

Every recommendation generated by CardioIQ must be fully decomposable. This distinguishes CardioIQ from opaque generative AI systems and ensures every output is transparent, reproducible, and auditable.

- Patient findings — specific biomarker values, clinical history, and demographic data
- Activated graph nodes — which ontology entities were engaged in the inference
- Traversed relationships — the specific edges followed during reasoning
- Evidence supporting each relationship — PMID, DOI, guideline reference, evidence class
- Applicable guideline recommendations — specific class and level of evidence
- Confidence score — graph-assigned weight for this inference pathway
- Human-readable explanation — plain-language clinical rationale

Regulatory Principle: CardioIQ adheres to the principle that AI in clinical decision support must be auditable. No recommendation may be generated without a traceable, evidence-supported inference path.

09 Update & Validation Strategy

Continuous Updates

The Knowledge Graph evolves through:

- New international guidelines
- Landmark RCTs and meta-analyses
- Regulatory label changes
- Expert consensus
- Version-controlled ontology revisions

All updates undergo scientific review before publication.

Rigorous Validation

Ontology Validation	Structural consistency and completeness of all nodes and edges
Clinical Consistency	Verification that all relationships reflect accepted pathophysiology
Evidence Traceability	Every edge linked to PMID, DOI, or named guideline document
Rule Conflict Detection	Automated detection of contradictory or redundant inference rules
Explainability Verification	All outputs decomposable into patient findings and graph paths
Physician Acceptance Testing	Clinical review by board-certified preventive cardiologists
Regression Testing	Automated re-validation following each knowledge base update
External Expert Review	Periodic review by independent lipidology and cardiology experts

10 API & Integration

CardioIQ exposes versioned, deterministic, auditable APIs. Designed for integration into EHRs, research platforms, and digital health applications.

Clinical Reasoning	Patient subgraph generation and inference execution
Guideline Mapping	Patient-specific guideline adherence assessment
Evidence Retrieval	Structured retrieval of supporting evidence for any edge
Drug Recommendation	Evidence-graded therapeutic recommendation support
Missing Data Detection	Identification of clinically required missing investigations
Patient Subgraph Generation	Dynamic personalised knowledge subgraph creation
Clinical Report Generation	Structured clinical intelligence report compilation

11 Future Roadmap

The architecture supports seamless expansion into:

• Polygenic Risk Scores (PRS) • Pharmacogenomics • Wearables (HR, activity, sleep) • Continuous Glucose Monitoring (CGM) • Digital biomarkers	• Retinal imaging AI • Echocardiography & CCTA AI • Multi-omics (proteomics, metabolomics, lipidomics) • Social determinants of health • Healthy Longevity analytics (biological age, aging biomarkers)
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Conclusion

CardiolQ Knowledge Graph™ redefines how cardiovascular knowledge is structured, reasoned, and applied. By combining semantic precision, evidence anchoring, and explainable AI, CardiolQ empowers clinicians to:

- Detect residual risk earlier — before it manifests as a clinical event
- Personalize therapy with confidence — grounded in current guidelines
- Align every decision with current ESC/EAS, ACC/AHA, ADA, and KDIGO recommendations
- Trust the "why" behind every recommendation — with full inference traceability

CardiolQ Knowledge Graph™ — semantic precision. Evidence anchored. Explainable AI. Designed for the future of precision cardiovascular care. [cardioiq.health](#)

Appendix A Example Patient Subgraph

58-year-old male · Post-NSTEMI/PCI (3 years) · T2DM (6 years) · CKD G3a · On: Atorvastatin, Aspirin, Beta-blocker, ACEi, Metformin

Finding	Graph Inference	Status
ApoB 104 mg/dL	Above very-high-risk target (<65 mg/dL) — 60% above goal	CRITICAL
Lp(a) 138 nmol/L	Significantly elevated — genetically mediated independent risk	CRITICAL
LDL-C 82 mg/dL	Above very-high-risk target (<1.4 mmol/L) despite statin	ABOVE TARGET
CKD G3a	Amplifies ASCVD risk — modifies treatment strategy	AMPLIFIER
hsCRP 3.8 mg/L	Elevated inflammatory residual risk — consider colchicine	ELEVATED
Urine ACR	MISSING — mandatory annual check per KDIGO 2022	MISSING

Risk: VERY HIGH (ESC/EAS 2019) | **ApoB Target:** <65 mg/dL | **Priority:** Add ezetimibe + SGLT2 inhibitor | **Follow-up:** 6–12 weeks

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Disclaimer: This white paper describes the CardioIQ Knowledge Graph™ architecture as of Version 1.0 (July 2026). CardioIQ is a clinical decision support tool intended for qualified healthcare professionals. It does not constitute a medical device, does not provide medical diagnoses, and does not replace the clinical judgment of a licensed physician.